

Predicting Corporate Bond YTM Changes Using Machine Learning

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1 Objective

The objective of this project is to predict corporate bond yield to maturity (YTM) changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators.

By applying various machine learning regression models such as ElasticNet, Boosting, and Multi-layer Perceptron, we aim to improve prediction accuracy compared to traditional linear regression models.

In addition, we used the change direction of YTM movements as a classification target since there are many outliers in the YTM change values that may affect regression model performance. We applied classification models such as Logistic Regression, ElasticNet Classifier, and Boosting Classifier to predict the direction of YTM changes.

2 Methodology

2.1 Data Description

We utilized data from Wharton Research Data Services (WRDS) to construct our dataset. The data frequency is monthly, and the data range is from July 2002 to February 2025.

We have a total of 50 features from four categories: bond-level features, firm-level features, macroeconomic features, and industry features. The details of the predictors are listed in the appendix.

The target variable is the corporate bond yield to maturity (YTM) change, defined as the difference in YTM over a one-month horizon:

$$\Delta\text{YTM}_i = \text{YTM}_{i,t} - \text{YTM}_{i,t-1} \quad (1)$$

where $\text{YTM}_{i,t}$ is the yield to maturity of bond i at the end of month t .

For classification, the target variable is defined as:

$$\text{YTM_Direction}_i = \begin{cases} \text{up,} & \text{if } \Delta\text{YTM}_i > 1e^{-6} \\ \text{neutral,} & \text{if } \Delta\text{YTM}_i \approx 0 \\ \text{down,} & \text{if } \Delta\text{YTM}_i < -1e^{-6} \end{cases} \quad (2)$$

The total number of observations after merging all datasets is approximately 854,769, covering 12,117 unique bonds over the sample period.

The data descriptive statistics of the yield to maturity (YTM) changes are as follows:

	count	mean	std	min	25%	50%	75%	max
Value	854769.00	-0.000005	0.01444	-0.9808	-0.00184	-0.00003	0.00171	0.97903

Table 1: Descriptive statistics (transposed to one row).

The number of observations for each YTM change direction category is as follows:

	down	up	neutral
Count	435398	399205	20166

Table 2: Class distribution (transposed).

2.2 Feature Construction

After preparing the bond-level, firm-level, sector-level, and macroeconomic datasets, we align all information at a monthly frequency and construct the final panel used for modeling. This stage focuses on dataset merging, temporal alignment, lagging to avoid look-ahead bias, missing-value handling, and feature normalization.

1. **Merging Procedure** We combine the datasets using index columns: **CUSIP** (bond identifier) and **Date** (end-of-month date).
2. **Missing-Value Handling** After merging, missing values are handled according to variable type:

- **Categorical Variables:** Credit rating dummies (AAA...D), Rating transition indicators, and Compustat data-status flag (costat). Here we use cross-sectionally monthly median to impute missing values, preserving the categorical distribution across firms each month.
- **Numeric Variables:** We also use cross-sectionally monthly median to impute missing values for numeric variables, then for remaining NaNs, we set them to zero.

3. Normalization

- **Rank-Normalized Features:** Includes stock features, fundamentals, derived ratios, growth variables, and bond-level numeric predictors. Each month, variables are transformed using:

$$\text{scaled_rank} = 2 \times \frac{\text{rank}}{N} - 1 \quad (3)$$

producing values in $[-1, 1]$. For binary variables (e.g., upgrade/downgrade), ranks are assigned such that 1 maps to 1 and 0 maps to -1 .

- **Not Normalized (kept in raw form):** Includes macroeconomic indicators and yield-curve variables (e.g., `sp500_ret`, `gdp_gr`, `cpi_infl`, `ir3m_chg`, `ir10y_chg`, `vix_chg`, `s3m`, `term_spread`). These variables are identical across all firms in a given month; rank-normalizing them collapses them into constants, eliminating their informational content. Therefore, they are kept in raw form to preserve meaningful macro signals.

2.3 Data Splitting

We used a rolling window forecasting approach. At each year, we used a time-series split to create training, validation, and test sets. The training set consisted of 10 years of data, the validation set and test set each consisted of 1 year of data. At each year, we tuned hyperparameters based on training set and validation set, then forecast target variables in the test set.

When tuning is not involved or cross-validation is adopted, the validation set was combined with the training set to train the final model. For example, to predict YTM changes in 2013, we used data from 2002 to 2011 as the training set, data in 2012 as the validation set, and data in 2013 as the test set.

2.4 Modeling Framework

2.4.1 Regression Models

We implemented and compared the following machine learning models for the regression task:

Model	Description
Linear Regression	Baseline linear model
ElasticNet	Combination of L1 and L2 regularization. We tuned alpha and l1_ratio
Boosting Regressor(LightGBM)	Ensemble of weak learners
Stacking Regressor	Ensemble of multiple models using linear regression as meta-model
Multi-layer Perceptron	Feedforward neural network

For the Multilayer Perceptron (MLP), we use the same rank-normalized predictors as in the other regression models (bond-level, firm-level, and sector-level features), together with the raw macro variables. The MLP is implemented as a small fully connected feed-forward network with ReLU activation functions and a single linear output node, trained with mean squared error loss and the Adam optimizer. For each rolling window, we re-estimate the MLP on the corresponding training period and evaluate it on the test year, using an internal hold-out split within the training window for early stopping. Thus, the MLP is directly comparable to the other regression models in terms of the rolling-window evaluation design, although it is considerably more computationally intensive and we keep the network architecture relatively small to control training time. The MLP is not part of the stacked regressor due to its separate training procedure and high computational cost.

Notice that in each load, we normalized the macroeconomic variables based on the training set statistics. That’s because these variables are identical across all firms in a given month, normalizing them using the entire dataset would lead to data leakage and normalizing them monthly would collapse them into constants, eliminating their informational content.

2.4.2 Classification Models

For classification task, we implemented and compared the following models:

Model	Description
Logistic Regression	Linear classification model
ElasticNet Classifier	Combination of L1 and L2 regularization. We tuned alpha and l1_ratio
Boosting Classifier (LightGBM)	Ensemble of weak learners. We tuned num_leaves and learning_rate hy
Stacking Classifier	Ensemble of multiple models using logistic regression as meta-model

2.5 Model Evaluation

We used a rolling window evaluation approach and calculated performance metrics on the test set for each year from 2012 to 2024. Then, the metrics were averaged to assess overall model performance.

For regression models, performance metrics included: MSE, MAE, MedAE, and R^2

For classification models, performance metrics included: Accuracy, Precision, and Recall

3 Results and Discussion

3.1 Regression Results

The table below summarizes the performance of different regression models on the test set:

Model	MSE	MAE	MedAE	R^2
ElasticNet	0.000097	0.003107	0.001900	-0.019272
Boosting Regressor(LightGBM)	0.000089	0.003069	0.001620	-0.109083
Linear Regression	0.000094	0.003226	0.002065	0.001753
Stacking Regressor	0.000086	0.002959	0.001591	-0.008823
Multilayer Perceptron	0.000103	0.003265	0.001792	0.232484

The Elastic Net Regression model did not significantly outperform the Linear Regression model. The Boosting Regressor (LightGBM) and the Stacking Regressor showed improved performance over linear models, with lower MSE and MAE values.

The multilayer perceptron (MLP) underperforms all other models in the rolling experiment, with noticeably higher MSE/MAE. This suggests that, given our current feature set

and simple network architecture, a feed-forward neural network does not extract additional predictive structure beyond what linear models and tree-based ensembles already capture, and may even be more prone to fitting noise.

3.2 Classification Results

The table below summarizes the performance of classification models on the test set:

Model	Accuracy	Precision	Recall
Elastic Net Classifier	0.687915	0.686969	0.687915
Boosting Classifier (LightGBM)	0.713470	0.719949	0.713470
Logistic Regression	0.687858	0.687190	0.687858
Stacking Classifier	0.713615	0.720544	0.713615

In classification tasks, the Boosting Classifier (LightGBM) and Stacking Classifier outperformed the logistic models.

4 Conclusion

In this project, we explored various machine learning models to predict corporate bond yield to maturity changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators. Our findings indicate that ensemble methods like LightGBM outperform traditional linear models in both regression and classification tasks. Regularization techniques such as ElasticNet did not yield significant improvements over linear models in this context. Stacking models didn't provide substantial gains over the best individual models.

A Appendix

A.1 A.1 Implementation Details

1. Software and Libraries

- CPU: MacBook Pro (Apple M4 Pro)
- Programming Language: Python 3.13
- Libraries: pandas, numpy, scikit-learn, lightgbm, tensorflow/keras
- Code Repository: Github https://github.com/QinqinAndMacaulayCat/ml_project/tree/main

2. Computation Time For Each Model (Approximate)

Model	Tuning	Training/Prediction
Linear Regression	-	3.9 seconds
Elastic Net Regression	-	1 min 10 sec
LGBM Regressor	31 min 41.4 sec	58.3 seconds
Stacked Regressor	-	6 min 50 sec
Logistic Regression	-	1 min 30 sec
Elastic Net Classification	83 min 59.8 sec	23 min 40.7 sec
Boosting Classifier	114 min 48.8 sec	1 min 27.9 sec
Stacked Classifier	-	107 min 11.9 sec
Multilayer Perceptron	-	3 min 16 sec

AI Tools Used

- ChatGPT for drafting and refining report text
- GitHub Copilot for code suggestions and completions

A.2 B.1 Predictor List

1. Key Columns

Field Name	Description
CUSIP	Committee on Uniform Securities Identification Procedures. Unique 9-digit identifier for each bond.
PERMNO	Permanent Number. Unique firm-level identifier from CRSP.
Date	End of Month Date.

2. Bond-level features

Field Name	Description
Time to Maturity (<code>tmt</code>)	The remaining time to maturity of the bond in years.
Coupon Rate (<code>coupon</code>)	The annual coupon rate of the bond.
<code>t_spread</code>	The weighted average bid-ask spread of the bond which measures liquidity.
<code>return_eom</code>	Bond return of last month.
Credit Ratings	One-hot encoded variables for ratings from AAA to D.
Rating Change Indicators	Binary variables (<code>upgrade</code> , <code>downgrade</code>).

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
tmt	3.56e+6	6.35	8.67	0.00	1.61	3.54	6.89	102.13
coupon	3.56e+6	0.03	0.03	0.00	0.00	0.03	0.06	0.34
t_spread	1.79e+6	0.01	0.01	0.00	0.00	0.00	0.01	2.00
yield	3.39e+6	0.01	0.13	-1.00	0.00	0.03	0.05	1.00
ret_eom	3.42e+6	0.01	1.18	-1.00	0.00	0.00	0.01	2090.0
rating_A	3.42e+6	0.20	0.40	0.00	0.00	0.00	0.00	1.00
rating_AA	3.42e+6	0.04	0.21	0.00	0.00	0.00	0.00	1.00
rating_AAA	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_B	3.42e+6	0.03	0.16	0.00	0.00	0.00	0.00	1.00
rating_BB	3.42e+6	0.04	0.20	0.00	0.00	0.00	0.00	1.00
rating_BBB	3.42e+6	0.23	0.42	0.00	0.00	0.00	0.00	1.00
rating_C	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
rating_CC	3.42e+6	0.00	0.03	0.00	0.00	0.00	0.00	1.00
rating_CCC	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_D	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
upgrade	3.42e+6	0.00	0.04	0.00	0.00	0.00	0.00	1.00
downgrade	3.42e+6	0.00	0.05	0.00	0.00	0.00	0.00	1.00

3. Firm-level features

We incorporate firm-level information from both daily stock data (CRSP) and quarterly accounting data (Compustat).

Table 3: Description of Firm-level Features

Variable	Description
<i>Stock Market Features</i>	
ret	Monthly stock return accumulated from daily returns.
vol	Monthly return volatility, calculated as the standard deviation of daily log returns.
dvol	Dollar trading volume, calculated as price multiplied by shares traded.
turnover	Share turnover ratio, defined as shares traded divided by shares outstanding.
bidask	Bid-ask spread based on daily closing bid and ask quotes, measuring liquidity cost.
numtrades	Total number of trades executed in the month.
mktcap	Month-end market capitalization (Price $\times$ Shares Outstanding).
price_mean	The average daily closing stock price over the month.
<i>Accounting-Based Derived Indicators</i>	
log_atq	Natural logarithm of total assets, used as a proxy for firm size.
lev_total	Total leverage ratio, calculated as total liabilities divided by total assets.
equity_ratio	Equity-to-assets ratio, calculated as total common equity divided by total assets.
roa	Return on Assets (ROA), defined as net income divided by total assets.
profit_margin	Net profit margin, calculated as net income divided by total revenue.
int_coverage	Interest coverage ratio, defined as operating income divided by interest expense.
mkt_cap	Accounting-based market capitalization.
market_to_book	Market-to-book ratio, calculated as market capitalization divided by book equity.
<i>Growth Indicators (QoQ)</i>	
atq_growth	Quarter-over-quarter growth rate of total assets.
revtq_growth	Quarter-over-quarter growth rate of total revenue.
niq_growth	Quarter-over-quarter growth rate of net income.

Variable	Mean	Std	Min	25%	50%	75%	Max
stock_ret	0.00081	0.1162	-0.3446	-0.052	0.00206	0.05164	0.3881
stock_vol	0.02420	0.01729	0	0.01106	0.01978	0.03373	0.07639
dvol	4.720e8	1.338e9	0	3.564e6	2.814e7	2.269e8	9.083e9
price_mean	27.89	33.39	0.4247	7.503	17.41	34.67	202.7
turnover	10.55	20.04	0.1046	2.171	5.032	10.39	151.0
bidask	0.03700	0.03154	0.00136	0.01481	0.02762	0.04922	0.1571
numtrades	2.353e4	7.754e4	0	0	0	5060	5.375e5
stock_mktcap	2.558e6	7.306e6	2612	6.575e4	2.860e5	1.336e6	5.040e7
log_atq	5.966	2.271	1.059	4.265	5.935	7.575	11.88
lev_total	0.5453	0.2667	0.04291	0.3332	0.5433	0.7454	1.437
equity_ratio	0.4682	0.2518	0.03758	0.2655	0.4584	0.6722	0.9650
roa	-0.011	0.06202	-0.4176	-0.001	0.00419	0.01634	0.1046
profit_margin	-0.067	0.5065	-4.911	-0.032	0.03624	0.09573	0.8525
int_coverage	7.883	10.21	-9.713	2.324	4.903	9.881	78.47
mkt_cap	3631	2.366e4	0.0150	55.60	249.9	1257	4.330e6
m2b	2.865	3.123	0.2618	1.144	1.846	3.275	26.64
atq_growth	0.0306	0.2174	-1.000	-0.023	0.00585	0.04315	4.966
revtq_growth	0.0594	0.3745	-1.000	-0.058	0.01596	0.1119	5.000
niq_growth	-0.111	1.158	-5.000	-0.538	-0.033	0.2378	5.000

4. Macroeconomic features

Table 4: Description of Firm-level Features

Variable	Description
<i>Stock Market Features</i>	
ret	Monthly stock return accumulated from daily returns.
vol	Monthly return volatility, calculated as the standard deviation of daily log returns.
dvol	Dollar trading volume, calculated as price multiplied by shares traded.
turnover	Share turnover ratio, defined as shares traded divided by shares outstanding.
bidask	Bid-ask spread based on daily closing bid and ask quotes, measuring liquidity cost.
numtrades	Total number of trades executed in the month.
mktcap	Month-end market capitalization (Price $\times$ Shares Outstanding).
price_mean	The average daily closing stock price over the month.
<i>Accounting-Based Derived Indicators</i>	
log_atq	Natural logarithm of total assets, used as a proxy for firm size.
lev_total	Total leverage ratio, calculated as total liabilities divided by total assets.
equity_ratio	Equity-to-assets ratio, calculated as total common equity divided by total assets.
roa	Return on Assets (ROA), defined as net income divided by total assets.
profit_margin	Net profit margin, calculated as net income divided by total revenue.
int_coverage	Interest coverage ratio, defined as operating income divided by interest expense.
mkt_cap	Accounting-based market capitalization.
market_to_book	Market-to-book ratio, calculated as market capitalization divided by book equity.
<i>Growth Indicators (QoQ)</i>	
atq_growth	Quarter-over-quarter growth rate of total assets.
revtq_growth	Quarter-over-quarter growth rate of total revenue.
niq_growth	Quarter-over-quarter growth rate of net income.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
SP500	335	2185.71	1418.34	735.09	1191.41	1454.60	2772.33	6840.20
IR3M	335	2.07	2.00	0.00	0.12	1.51	4.21	6.15
IR10Y	335	3.44	1.37	0.54	2.30	3.53	4.49	6.67
VIX	335	20.29	7.82	9.51	14.45	18.43	24.44	59.89
GDP	328	17764.9	2911.2	12703.7	15670.8	17035.1	20070.6	23770.9
CPI	333	227.69	42.89	162.00	191.70	227.17	252.56	324.37

5. Industry features

Table 5: Description of Industry Features

Field Name	Description
price	The monthly closing price of the sector ETF corresponding to the firm’s Global Industry Classification Standard (GICS) sector.
return	The monthly return of the sector ETF, used as a proxy for sector-specific market performance and shocks.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Price	3128	44.03	39.40	4.57	18.23	29.43	54.97	300.68
Return	3117	0.008	0.055	-0.3437	-0.0217	0.0110	0.0390	0.3076

A.2.1 B.1 Classification Confusion Matrix

1. Logistic Classification

Actual \ Pred	Up	Down	Neutral
Up	155810	103344	200
Down	52584	207780	286
Neutral	4410	6969	79

2. Elastic Net Classification

Actual \ Pred	Up	Down	Neutral
Up	155781	103385	188
Down	52524	207861	265
Neutral	4418	6975	65

3. LGBM Classifier

Actual \ Pred	Up	Down	Neutral
Up	176082	82742	530
Down	62480	197594	576
Neutral	2382	3005	6071

4. Stacked Classifier

Actual \ Pred	Up	Down	Neutral
Up	178684	80475	195
Down	64062	196423	165
Neutral	3217	4005	4236